

Electrolysis-Free WATER-LEVEL ALARM

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In most water-level monitors and alarms, corrosion of the probes makes the systems unusable in a few months. This water-level alarm uses a phase-locked loop (PLL) circuit to generate a particular frequency at its probes to avoid electrolysis of the water. The circuit uses two NE567 PLL ICs.

Circuit and working

Circuit diagram of the electrolysis-free water-level alarm is shown in Fig. 1. It is built around two NE567 ICs (IC1 and IC2), npn transistor BC547 (T1), pnp transistor BC557

(T2) and piezo buzzer (PZ1), along with a few other components.

IC1 generates around 500Hz signals through its voltage-controlled oscillator, which is applied to first probe P1. When P1 gets shorted with second probe P2 via water, IC2 decodes this signal and activates buzzer PZ1. When a frequency of around 500Hz is received by IC2, its output pin 8 goes low. This makes pnp transistor T2 to conduct, which, in turn, makes LED1 glow and PZ1 beep.

As the signal is passed through water, there is no electrolysis and, hence, corrosion of the probes (P1

PARTS LIST

Semiconductors:

- IC1, IC2 - NE567 PLL
- T1 - BC547 npn transistor
- T2 - BC557 pnp transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R5 - 10-kilo-ohm
- R2 - 470-kilo-ohm
- R3, R6 - 1-kilo-ohm
- R4 - 100-kilo-ohm
- R7 - 470-ohm

Capacitors:

- C1, C2, C7 - 10 μ F, 25V electrolytic
- C3, C9 - 0.22 μ F ceramic disk
- C4, C5 - 0.47 μ F ceramic disk
- C6 - 0.01 μ F ceramic disk
- C8 - 4.7 μ F, 25V electrolytic
- C10, C12 - 220 μ F, 25V electrolytic
- C11 - 0.1 μ F ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- S1 - On/off switch
- PZ1 - Piezo buzzer
- P1, P2 - Steel sensor probes
- 9V battery/9V DC power supply

and P2) does not occur.

The circuit works off any 5V DC to 9V DC power supply.

Construction and testing

An actual-size, single-side PCB layout for the electrolysis-free water-level alarm is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a plastic box and connect the sensors using a two-wire cable to the PCB.

Install the probes in the overhead tank in such a way that when the tank gets filled, both probes get shorted through water. **EFY**

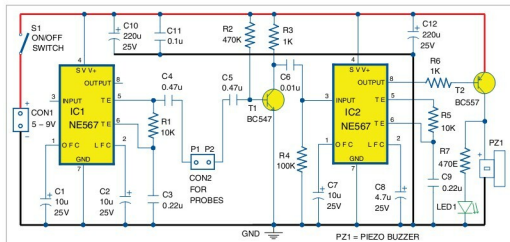


Fig. 1: Circuit diagram of the electrolysis-free water-level alarm

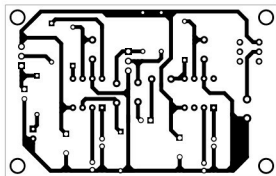


Fig. 2: Actual-size PCB layout of the electrolysis-free water-level alarm

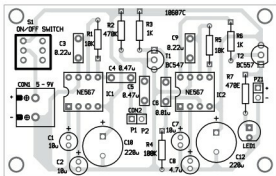
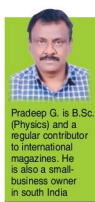


Fig. 3: Component layout of the PCB



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